

Class 8: Unsupervised Learning Analysis of Human Breast Cancer Cells

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Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today we will analyze a biopsy data-set from fine needle aspiration (FNA) of breast mass.

Data Import

The data is made available as a CSV file for download. We can read this in using 'read.csv()':

```
fna.data <- "WisconsinCancer.csv"
wisc.df <- read.csv(fna.data, row.names=1)
```

Make sure we remove or exclude the 'diagnosis' column from the data set that we use for further analysis - this is the expert diagnosis as either M or B.

```
wisc.data <- wisc.df[,-1]
diagnosis <- as.factor(wisc.df$diagnosis)
```

Exploratory data analysis

Q1. How many observations are in this dataset?

```
nrow(wisc.data)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis?

```
table(wisc.df$diagnosis)
```

```
  B    M
357 212
```

212 observations have a malignant diagnosis.

Q3. How many variables/features in the data set are suffixed with `_mean`?

We can use the `'grep()'` function to help us here:

```
colnames(wisc.data)
```

```
[1] "radius_mean"      "texture_mean"
[3] "perimeter_mean"   "area_mean"
[5] "smoothness_mean"  "compactness_mean"
[7] "concavity_mean"    "concave.points_mean"
[9] "symmetry_mean"     "fractal_dimension_mean"
[11] "radius_se"         "texture_se"
[13] "perimeter_se"      "area_se"
[15] "smoothness_se"     "compactness_se"
[17] "concavity_se"      "concave.points_se"
[19] "symmetry_se"       "fractal_dimension_se"
[21] "radius_worst"      "texture_worst"
[23] "perimeter_worst"   "area_worst"
[25] "smoothness_worst"  "compactness_worst"
[27] "concavity_worst"    "concave.points_worst"
[29] "symmetry_worst"     "fractal_dimension_worst"
```

```
length(grep("_mean", colnames(wisc.data), value=T))
```

```
[1] 10
```

Principal Component Analysis

We need to scale our data before PCA with the ‘scale=TRUE’ argument to ‘prcomp()’.

```
wisc.pr <- prcomp(wisc.data, scale=TRUE)
summary(wisc.pr)
```

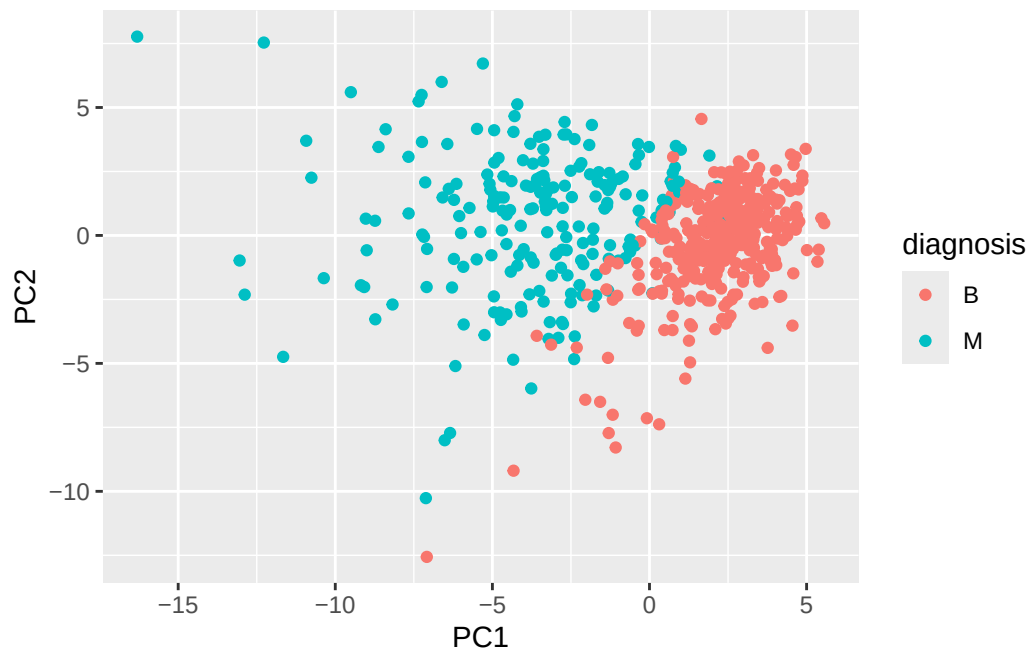
Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Let’s see the “PC score plot”

```
library(ggplot2)

ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



```
colMeans(wisc.data)
```

radius_mean	texture_mean	perimeter_mean
1.412729e+01	1.928965e+01	9.196903e+01
area_mean	smoothness_mean	compactness_mean
6.548891e+02	9.636028e-02	1.043410e-01
concavity_mean	concave.points_mean	symmetry_mean
8.879932e-02	4.891915e-02	1.811619e-01
fractal_dimension_mean	radius_se	texture_se
6.279761e-02	4.051721e-01	1.216853e+00
perimeter_se	area_se	smoothness_se
2.866059e+00	4.033708e+01	7.040979e-03
compactness_se	concavity_se	concave.points_se
2.547814e-02	3.189372e-02	1.179614e-02
symmetry_se	fractal_dimension_se	radius_worst
2.054230e-02	3.794904e-03	1.626919e+01
texture_worst	perimeter_worst	area_worst
2.567722e+01	1.072612e+02	8.805831e+02
smoothness_worst	compactness_worst	concavity_worst
1.323686e-01	2.542650e-01	2.721885e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
1.146062e-01	2.900756e-01	8.394582e-02

```
apply(wisc.data,2,sd)
```

radius_mean	texture_mean	perimeter_mean
3.524049e+00	4.301036e+00	2.429898e+01
area_mean	smoothness_mean	compactness_mean
3.519141e+02	1.406413e-02	5.281276e-02
concavity_mean	concave.points_mean	symmetry_mean
7.971981e-02	3.880284e-02	2.741428e-02
fractal_dimension_mean	radius_se	texture_se
7.060363e-03	2.773127e-01	5.516484e-01
perimeter_se	area_se	smoothness_se
2.021855e+00	4.549101e+01	3.002518e-03
compactness_se	concavity_se	concave.points_se
1.790818e-02	3.018606e-02	6.170285e-03
symmetry_se	fractal_dimension_se	radius_worst
8.266372e-03	2.646071e-03	4.833242e+00
texture_worst	perimeter_worst	area_worst
6.146258e+00	3.360254e+01	5.693570e+02
smoothness_worst	compactness_worst	concavity_worst
2.283243e-02	1.573365e-01	2.086243e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
6.573234e-02	6.186747e-02	1.806127e-02

```
wisc.pr <- prcomp(wisc.data, scale. = TRUE)
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
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Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987

Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

0.4427

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

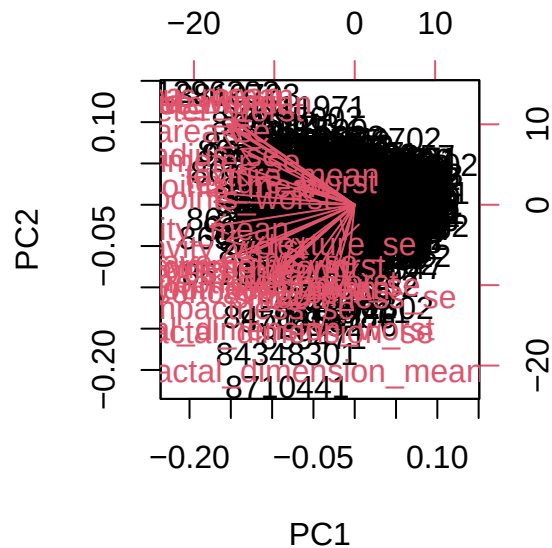
3

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

7

Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

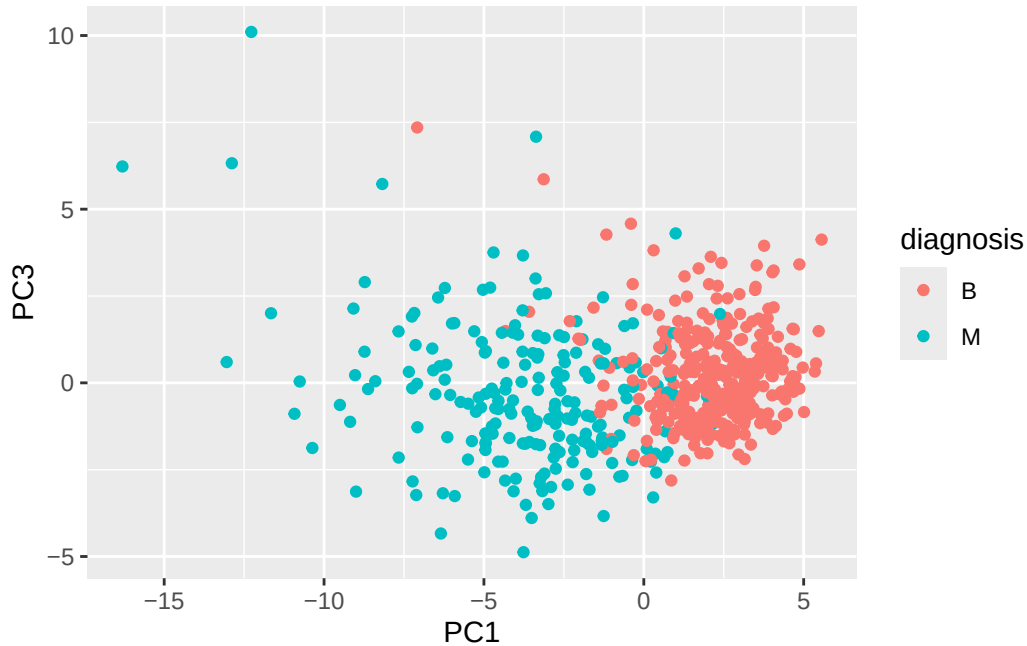
```
biplot(wisc.pr)
```



Everything is on top of each other, so it's very difficult to understand.

Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
ggplot(wisc.pr$x) +  
  aes(PC1, PC3, col=diagnosis) +  
  geom_point()
```



There's a separation between the red (benign) and blue (malignant) samples.

Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation[,1]
```

radius_mean	texture_mean	perimeter_mean
-0.21890244	-0.10372458	-0.22753729
area_mean	smoothness_mean	compactness_mean
-0.22099499	-0.14258969	-0.23928535
concavity_mean	concave.points_mean	symmetry_mean

-0.25840048	-0.26085376	-0.13816696
fractal_dimension_mean	radius_se	texture_se
-0.06436335	-0.20597878	-0.01742803
perimeter_se	area_se	smoothness_se
-0.21132592	-0.20286964	-0.01453145
compactness_se	concavity_se	concave.points_se
-0.17039345	-0.15358979	-0.18341740
symmetry_se	fractal_dimension_se	radius_worst
-0.04249842	-0.10256832	-0.22799663
texture_worst	perimeter_worst	area_worst
-0.10446933	-0.23663968	-0.22487053
smoothness_worst	compactness_worst	concavity_worst
-0.12795256	-0.21009588	-0.22876753
concave.points_worst	symmetry_worst	fractal_dimension_worst
-0.25088597	-0.12290456	-0.13178394

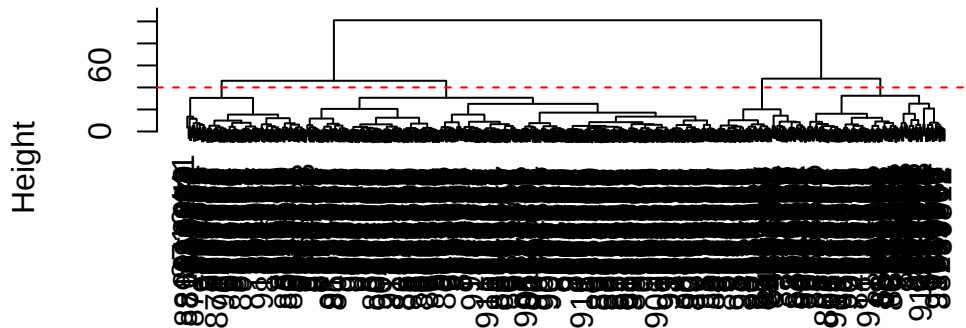
It is -0.26085376. No other features have larger contributions.

Hierarchical clustering

Q10. Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

```
d <- dist(wisc.pr$x[,1:4])
wisc.pr.hclust <- hclust(d, method="ward.D2")
plot(wisc.pr.hclust)
abline(h=40, col="red", lty=2)
```


Cluster Dendrogram



d
hclust (*, "ward.D2")

The model has 4 clusters at $h=40$.

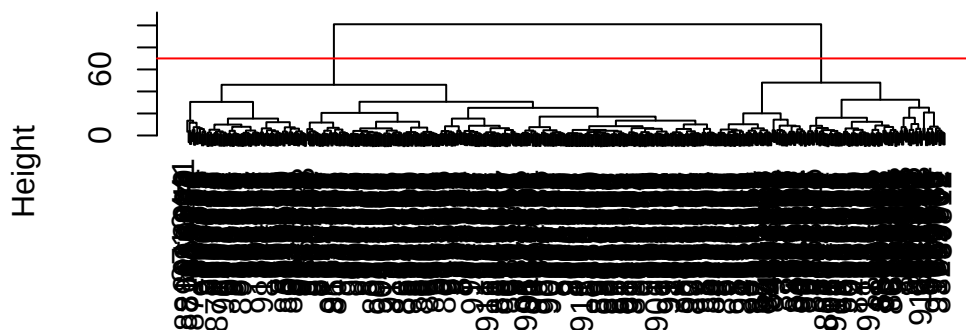
Q12. Which method gives your favorite results for the same data.dist dataset?
Explain your reasoning.

The “ward.D2” method gives my favorite results because the separation between groups is cleaner and easier to see.

Combining Methods

```
d <- dist(wisc.pr$x[,1:4])
wisc.pr.hclust <- hclust(d, method="ward.D2")
plot(wisc.pr.hclust)
abline(h=70, col="red")
```

Cluster Dendrogram



d
hclust (*, "ward.D2")

```
grps <- cutree(wisc.pr.hclust, h=70)
table(grps)
```

```
grps
  1  2
171 398
```

How does this clustering 'grps' correspond to the expert 'diagnosis'?

```
table(diagnosis)
```

```
diagnosis
  B  M
357 212
```

Q13. How well does the newly created hclust model with two clusters separate out the two "M" and "B" diagnoses?

```
table(diagnosis, grps)
```

```

      grps
diagnosis 1  2
B         6 351
M        165 47

```

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.hclust.clusters` and `wisc.pr.hclust.clusters`) with the vector containing the actual diagnoses.

```
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, h = 40)
```

```

d2 <- dist(wisc.data)
wisc.hclust <- hclust(d2, method = "ward.D2")
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)

```

```
table(diagnosis, wisc.hclust.clusters)
```

```

      wisc.hclust.clusters
diagnosis 1  2  3  4
B         0 260 97  0
M        75  6 120 11

```

```
table(diagnosis, wisc.pr.hclust.clusters)
```

```

      wisc.pr.hclust.clusters
diagnosis 1  2  3  4
B         0  6 267 84
M       104 61 41  6

```

Prediction

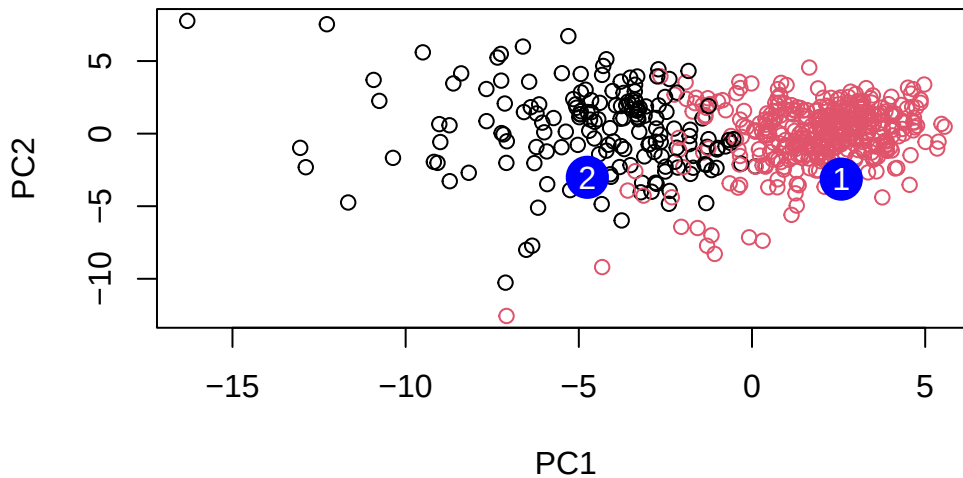
```

url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc

```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029
	PC15	PC16	PC17	PC18	PC19	PC20	
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216	
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500	
	PC21	PC22	PC23	PC24	PC25	PC26	
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581	
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238	
	PC27	PC28	PC29	PC30			
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029			
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820			

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

Patient 1 should be prioritized.